

Hypothesis test of conditional subsampling

HaiYing Wang

Department of Statistics, University of Connecticut

March 26, 2026

1 Hypothesis Test

We note that that the score function of IPW is

$$\psi_W(\boldsymbol{\theta}; \mathbf{x}, y, \delta) = \frac{\delta}{\pi(\mathbf{x}, y)} \nabla_{\boldsymbol{\theta}} \dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}),$$

and

$$\sum_{i=1}^N \psi_W(\hat{\boldsymbol{\theta}}_W; \mathbf{x}_i, y_i, \delta_i) = 0, \tag{1} \quad \text{\texttt{\{eq:score_W\}}}$$

where $\hat{\boldsymbol{\theta}}_W$ is the IPW estimator. Samewise, the score function of MSCLE is

$$\psi_S(\boldsymbol{\theta}; \mathbf{x}, y, \delta) = \delta [\nabla_{\boldsymbol{\theta}} \dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}) - \nabla_{\boldsymbol{\theta}} \log \bar{\pi}(\mathbf{x}, y)], \tag{2}$$

and

$$\sum_{i=1}^N \psi_S(\hat{\boldsymbol{\theta}}_S; \mathbf{x}_i, y_i, \delta_i) = 0, \tag{3} \quad \text{\texttt{\{eq:score_S\}}}$$

where $\hat{\boldsymbol{\theta}}_S$ is the MSCLE and $\bar{\pi}(\mathbf{x}, y) = \int f(y | \mathbf{x}, \boldsymbol{\theta}) \pi(\mathbf{x}, y)$. As it is mentioned by Wang & Kim (2022), the IPW estimator $\hat{\boldsymbol{\beta}}_W$ is consistent, because the conditional expectation of the selection indicator δ give the data equals the inclusion probability and therefore the expectation of the objective function for the IPW estimator is the same as that of the full data MLE. Yet, the MSCLE is estimated by the assumed model structure to derive the subsampled conditional likelihood, so if the subsampling probabilities also depend on the data then the MSCLE is not consistent to the same limit as the full data MLE when the assumed model structure is misspecified. That brings a thought that the distance between IPW estimator and MSCLE can be used as an indicator of model misspecification since this distance vanishes

asymptotically in the absence of misspecification, but generally doesn't vanish otherwise. Hausman-type test Hausman (1978), White (1982) is established to evaluate the difference between these two parameters, i.e., one is stable when no misspecification, while the other is more efficient yet not that stable. The misspecification indicator is $\hat{\boldsymbol{\theta}}_S - \hat{\boldsymbol{\theta}}_W$, so we investigate the asymptotic distribution of $\sqrt{N}(\hat{\boldsymbol{\theta}}_S - \hat{\boldsymbol{\theta}}_W)$. We anticipate that with appropriate regularity conditions, $\sqrt{N}(\hat{\boldsymbol{\theta}}_S - \hat{\boldsymbol{\theta}}_W)$ will be normally distributed asymptotically with mean zero in the absence of misspecification. Given a consistent estimator for the asymptotic covariance matrix, we can form an asymptotic chi-square statistic.

Assumption 1. (1) $\|\psi_{\mathcal{T}}(\boldsymbol{\theta}; \mathbf{x}, y, \delta)\|$ and $\|\nabla_{\boldsymbol{\theta}}\psi_{\mathcal{T}}(\boldsymbol{\theta}; \mathbf{x}, y, \delta)\|$ are integrable, where $\mathcal{T} = \{W, S, SM\}$ and $\|A\| = \text{tr}^{1/2}(A^{\top}A)$ is the norm for a vector or matrix A .

(2) The parameter space $\boldsymbol{\theta}$ is compact and the third order partial derivative of $\psi_{\mathcal{T}}(y | \mathbf{x}, \boldsymbol{\theta})$ and $\log \bar{\pi}(\mathbf{x}, y)$ with respect to any components of $\boldsymbol{\theta}$ is bounded in absolute value by an integrable function $B(\mathbf{x}; y)$ that does not depend on $\boldsymbol{\theta}$.

(3) The variance matrix of the two estimators are finite and positive definite.

(4) $\boldsymbol{\theta}_0$ is the unique solution to the population minimization problem $\min_{\boldsymbol{\theta} \in \Theta} \mathbf{E}[\dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta})]$.

(5) Assume that the weighted objection function $\ell_W(y | \mathbf{x}, \boldsymbol{\theta})$ is twice continuously differentiable and is bounded in absolute value by an integrable function $B(\mathbf{x}; y)$ that does not depend on $\boldsymbol{\theta}$ (Note: This may overlap with Assumption (2).)

(6) $\Sigma_W^{-1} - \Sigma_S^{-1}$ is positive semidefinite. However, according to the Theorem 2 in Wang & Kim (2022), $\Sigma_W^{-1} \geq \Sigma_S^{-1}$, hence this condition is automatically satisfied.

Besides, we also make an Assumption 2 that all the properties in Assumption 1 are satisfied for the misspecified model f^w .

Assumptions (1)–(3) provide the essential regularity conditions for the asymptotic properties of the MSCLE estimator. Assumptions (3)–(5) furnish the necessary regularity conditions for the asymptotic properties of the IPW estimator. Assumption (6) ensures the existence of the covariance matrix of $\hat{\boldsymbol{\theta}}_S - \hat{\boldsymbol{\theta}}_W$, which supports Lemma 1.

Lemma 1. Under the null hypothesis of no misspecification, we consider the test statistic

$$H_N = N(\hat{\boldsymbol{\theta}}_S - \hat{\boldsymbol{\theta}}_W)\hat{V}_H^{-1}(\hat{\boldsymbol{\theta}}_S - \hat{\boldsymbol{\theta}}_W)^{\top}.$$

We have $\lim_{N \rightarrow \infty} H_N \xrightarrow{d} \chi_p^2$, where p is the parameter dimension and $\hat{V}_H$ is the approximated estimation of covariate matrix of $\sqrt{N}(\hat{\boldsymbol{\theta}}_S - \hat{\boldsymbol{\theta}}_W)$ that $\lim_{N \rightarrow \infty} \hat{V}_H = \Sigma_W^{-1} - \Sigma_S^{-1}$. Σ_W and Σ_S are the inverse of asymptotic matrices of IPW estimator and MSCLE respectively.

48 *Proof.* Since $\hat{\boldsymbol{\theta}}_W$ and $\hat{\boldsymbol{\theta}}_S$ all have the asymptotic properties, then under the null hypothesis,
 49 it follows from Liu et al. (2016), Wang & Kim (2022) that

$$\lim_{N \rightarrow \infty} \sqrt{N} \hat{\boldsymbol{\theta}}_W \xrightarrow{d} \mathbf{N}(\boldsymbol{\theta}_0, \Sigma_W^{-1}),$$

50 and

$$\lim_{N \rightarrow \infty} \sqrt{N} \hat{\boldsymbol{\theta}}_S \xrightarrow{d} \mathbf{N}(\boldsymbol{\theta}_0, \Sigma_S^{-1}),$$

51 where $\boldsymbol{\theta}_0$ is the true parameter.

52 In this case, we derictly use the Hausman's Lemma Consider the difference estimator
 53 $D_N = \hat{\boldsymbol{\theta}}_S - \hat{\boldsymbol{\theta}}_W$, we first have $\lim_{N \rightarrow \infty} \sqrt{N} D_N \xrightarrow{d} \mathbf{N}(0, V_H)$, that the approximation estimate of
 54 V_H can be derived as

$$\hat{V}_H = \mathbb{V}(\sqrt{N}(\hat{\boldsymbol{\theta}}_S - \hat{\boldsymbol{\theta}}_W)) \xrightarrow{d} \Sigma_W^{-1} + \Sigma_S^{-1} - 2N\text{COV}(\hat{\boldsymbol{\theta}}_W, \hat{\boldsymbol{\theta}}_S).$$

55 Then we just need to calculate the limit form of $\text{COV}(\hat{\boldsymbol{\theta}}_W, \hat{\boldsymbol{\theta}}_S)$.

56 $\hat{\boldsymbol{\theta}}_W$ and $\hat{\boldsymbol{\theta}}_S$ are the solutions of (1) and (3), so we expense these two equations at the true
 57 value $\boldsymbol{\theta}_0$ and have

$$0 = \frac{1}{\sqrt{N}} \sum_{i=1}^N \psi_W(\boldsymbol{\theta}_0; \mathbf{x}_i, y_i, \delta_i) + \left[\frac{1}{N} \sum_{i=1}^N \nabla_{\boldsymbol{\theta}} \psi_W(\boldsymbol{\theta}_0; \mathbf{x}_i, y_i, \delta_i) \right] \sqrt{N}(\hat{\boldsymbol{\theta}}_W - \boldsymbol{\theta}_0) + o_P(1), \quad (4) \quad \{\text{eq:tay_psi}\}$$

$$0 = \frac{1}{\sqrt{N}} \sum_{i=1}^N \psi_S(\boldsymbol{\theta}_0; \mathbf{x}_i, y_i, \delta_i) + \left[\frac{1}{N} \sum_{i=1}^N \nabla_{\boldsymbol{\theta}} \psi_S(\boldsymbol{\theta}_0; \mathbf{x}_i, y_i, \delta_i) \right] \sqrt{N}(\hat{\boldsymbol{\theta}}_S - \boldsymbol{\theta}_0) + o_P(1). \quad (5) \quad \{\text{eq:tay_psi}\}$$

58 According to the Assumption 1(1), we assume that $A_W = \mathbb{E}[\nabla_{\boldsymbol{\theta}} \psi_W(\boldsymbol{\theta}_0; \mathbf{x}, y, \delta)]$ and $A_S =$
 59 $\mathbb{E}[\nabla_{\boldsymbol{\theta}} \psi_S(\boldsymbol{\theta}_0; \mathbf{x}, y, \delta)]$, with the large law of number, we have

$$\frac{1}{N} \sum_{i=1}^N \nabla_{\boldsymbol{\theta}} \psi_W(\boldsymbol{\theta}_0; \mathbf{x}_i, y_i, \delta_i) \xrightarrow{P} A_W,$$

60 and

$$\frac{1}{N} \sum_{i=1}^N \nabla_{\boldsymbol{\theta}} \psi_S(\boldsymbol{\theta}_0; \mathbf{x}_i, y_i, \delta_i) \xrightarrow{P} A_S.$$

61 Besides,

$$\left[\frac{1}{N} \sum_{i=1}^N \nabla_{\boldsymbol{\theta}} \psi_W(\boldsymbol{\theta}_0; \mathbf{x}_i, y_i, \delta_i) \right] \left[\frac{1}{N} \sum_{i=1}^N \nabla_{\boldsymbol{\theta}} \psi_S(\boldsymbol{\theta}_0; \mathbf{x}_i, y_i, \delta_i) \right] \xrightarrow{P} \mathbb{E}[\psi_W(\boldsymbol{\theta}_0; \mathbf{x}, y, \delta) \psi_S(\boldsymbol{\theta}_0; \mathbf{x}, y, \delta)^\top] = B_{WS},$$

62 then we have

$$B_{WS} = \mathbb{E} \left[\frac{\delta}{\pi(\mathbf{x}, y)} \nabla_{\boldsymbol{\theta}} \dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}_0) \left(\nabla_{\boldsymbol{\theta}} \dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}_0) - \nabla_{\boldsymbol{\theta}} \log \bar{\pi}(\mathbf{x}, y) \right)^\top \right]$$

$$= \mathbb{E}_{(\mathbf{x}, y)} \left[\mathbb{E} \left[\frac{\delta}{\pi(\mathbf{x}, y)} \nabla_{\boldsymbol{\theta}} \dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}_0) \left(\nabla_{\boldsymbol{\theta}} \dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}_0) - \nabla_{\boldsymbol{\theta}} \log \bar{\pi}(\mathbf{x}, y) \right)^{\top} \right] \mid (\mathbf{x}, y) \right]. \quad (6)$$

Since $\mathbb{E}[\delta | \mathbf{x}, y] = \pi(\mathbf{x}, y)$,

$$\begin{aligned} B_{WS} &= \mathbb{E}_{(\mathbf{x}, y)} \left[\nabla_{\boldsymbol{\theta}} \dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}_0) \left(\nabla_{\boldsymbol{\theta}} \dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}_0) - \nabla_{\boldsymbol{\theta}} \log \bar{\pi}(\mathbf{x}, y) \right)^{\top} \right] \\ &= \mathbb{E}_{(\mathbf{x}, y)} \left[\nabla_{\boldsymbol{\theta}} \dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}_0) \nabla_{\boldsymbol{\theta}} \dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}_0)^{\top} \right] - \mathbb{E}_{(\mathbf{x}, y)} \left[\nabla_{\boldsymbol{\theta}} \dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}_0) \nabla_{\boldsymbol{\theta}} \log \bar{\pi}(\mathbf{x}, y)^{\top} \right] \\ &= \mathbb{E}_{(\mathbf{x}, y)} \left[\nabla_{\boldsymbol{\theta}} \dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}_0) \nabla_{\boldsymbol{\theta}} \dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}_0)^{\top} \right] - \mathbb{E}_{(y)} \left[\mathbb{E}_{(\mathbf{x})} \left[\nabla_{\boldsymbol{\theta}} \dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}_0) \right] \nabla_{\boldsymbol{\theta}} \log \bar{\pi}(\mathbf{x}, y)^{\top} \right] \\ &= \mathbb{E}_{(\mathbf{x}, y)} \left[\nabla_{\boldsymbol{\theta}} \dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}_0) \nabla_{\boldsymbol{\theta}} \dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}_0)^{\top} \right], \end{aligned} \quad (7) \quad \{\mathbf{B}_{WS}\}$$

where we use $\mathbb{E}_{(\mathbf{x})} \left[\nabla_{\boldsymbol{\theta}} \dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}_0) \right] = 0$ in this case.

Besides, with the Assumption 1(1) and (5), and Eq. 7, we have

$$A_W = -\mathbb{E}[-\nabla_{\boldsymbol{\theta}} \psi_W(\boldsymbol{\theta}_0; \mathbf{x}, y, \delta)] = -\mathbb{E}_{(\mathbf{x}, y)} [-\nabla_{\boldsymbol{\theta}}^2 \ell_W(\boldsymbol{\theta}_0; \mathbf{x}, y, \delta)] = B_{WS},$$

while

$$A_S = \mathbb{E}[-\nabla_{\boldsymbol{\theta}}^2 \ell_S(\boldsymbol{\theta}_0; \mathbf{x}, y, \delta)] = \Sigma_S,$$

with the conclusion in Wang & Kim (2022) (A.12) that $-\frac{1}{N} \nabla_{\boldsymbol{\theta}}^2 \ell_S(\boldsymbol{\theta}_0; \mathbf{x}, y, \delta) = \Sigma_S + o_P(1)$.

Thus we have

$$\lim_{N \rightarrow \infty} \sqrt{N}(\hat{\boldsymbol{\theta}}_W - \boldsymbol{\theta}_0) = -A_W^{-1} \left(\frac{1}{\sqrt{N}} \sum_{i=1}^N \psi_W(\boldsymbol{\theta}_0; \mathbf{x}_i, y_i, \delta_i) \right) + o_P(1),$$

and

$$\lim_{N \rightarrow \infty} \sqrt{N}(\hat{\boldsymbol{\theta}}_S - \boldsymbol{\theta}_0) = -A_S^{-1} \left(\frac{1}{\sqrt{N}} \sum_{i=1}^N \psi_S(\boldsymbol{\theta}_0; \mathbf{x}_i, y_i, \delta_i) \right) + o_P(1).$$

Then

$$\lim_{N \rightarrow \infty} \mathbb{C}\mathbb{O}\mathbb{V}(\sqrt{N}\hat{\boldsymbol{\theta}}_W, \sqrt{N}\hat{\boldsymbol{\theta}}_S) = \lim_{N \rightarrow \infty} E[(\hat{\boldsymbol{\theta}}_W - \boldsymbol{\theta}_0)(\hat{\boldsymbol{\theta}}_S - \boldsymbol{\theta}_0)^{\top}] = A_W^{-1} B_{WS} A_S^{-1\top} = \Sigma_S^{-1}.$$

Recall that

$$\hat{V}_H \xrightarrow{P} \Sigma_W^{-1} + \Sigma_S^{-1} - 2N\mathbb{C}\mathbb{O}\mathbb{V}(\hat{\boldsymbol{\theta}}_W, \hat{\boldsymbol{\theta}}_S) \xrightarrow{P} \Sigma_W^{-1} - \Sigma_S^{-1}.$$

Thus based on the property of Wald-type statistics Van der Vaart (2000) and Slutsky's Theorem, the test statistic H_N follows a chi-square distribution χ_p^2 . $\square$

As a result, H_N can be used as a test statistic to decide whether the IPW or the MSCLE estimator should be chosen. The corresponding hypothesis test is formulated as follows:

$$H_0 : \mathbf{E}[\hat{\boldsymbol{\theta}}_S - \hat{\boldsymbol{\theta}}_W] = 0 \leftrightarrow H_1 : \mathbf{E}[\hat{\boldsymbol{\theta}}_S - \hat{\boldsymbol{\theta}}_W] \neq 0,$$

that null hypothesis H_0 represents that the model assumption $\dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta})$ is correct while the alternative hypothesis H_1 represents that the model is misspecified. We then compute the p -value $p_H = 1 - F_{\chi_p^2}(H_N)$ and compare it with the predetermined significance level α . If $p_H \leq \alpha$, we reject H_0 and choose the IPW estimator for more stable estimation; otherwise, we accept H_0 and select the MSCLE estimator for greater efficiency. Thus, we define a new combined estimator that incorporates both the IPW and MSCLE estimators via the hypothesis-test indicator:

$$\hat{\boldsymbol{\theta}}_N = \mathbf{I}\{H_N \geq c_\alpha\} \hat{\boldsymbol{\theta}}_W + \mathbf{I}\{H_N < c_\alpha\} \hat{\boldsymbol{\theta}}_S, \quad (8)$$

where c_α is the upper confidence limit of level α for the chi-squared test and $\mathbf{I}(\cdot)$ is the indicator function. We now study the property of the new estimator $\hat{\boldsymbol{\theta}}_N$, considering two separate cases. The situation should be discussed separately. First, if H_0 is accepted, when $n \rightarrow \infty$, we have $P(H_N > c_\alpha) \rightarrow \alpha$, then we have probability α to choose IPW estimator and probability $1 - \alpha$ to choose MSCLE. When the model is correctly specified or both the IPW and MSCLE estimators converge to the same full data estimator $\boldsymbol{\theta}_l$, $\hat{\boldsymbol{\theta}}_N$ is the unbiased estimator since it is the combination of two unbiased estimator

$$\mathbf{E}[\hat{\boldsymbol{\theta}}_N - \boldsymbol{\theta}_0] = \mathbf{E}[\mathbf{I}\{H_N \geq c_\alpha\}(\hat{\boldsymbol{\theta}}_W - \boldsymbol{\theta}_0)] + \mathbf{E}[\mathbf{I}\{H_N < c_\alpha\}(\hat{\boldsymbol{\theta}}_S - \boldsymbol{\theta}_0)] = 0.$$

Since

$$\sqrt{N}(\hat{\boldsymbol{\theta}}_W - \hat{\boldsymbol{\theta}}_S) \sim \mathbf{N}(0, \Sigma_S^{-1}).$$

and the sample space of $\{(\hat{\boldsymbol{\theta}}_W - \hat{\boldsymbol{\theta}}_S) \in \mathcal{R}^k : \mathbf{I}\{H_N \geq c_\alpha\} = 1\}$ should be a exterior of ellipsoid, eq (??) equals to 0. Thus, we have $\hat{\boldsymbol{\theta}}_N$ is an unbiased estimator.

When the model is misspecified, the score function of MSCLE would be a little different from Eq. 3 since we have to calculate the expectation $\bar{\pi}(\mathbf{x}, y) = \int f(y | \mathbf{x}, \boldsymbol{\theta}) \pi(\mathbf{x}, y)$, which depends on the assumption of ℓ . Thus, the score function of MSCLE under model misspecification is

$$\psi_{SM}(\boldsymbol{\theta}; \mathbf{x}_i, y_i, \delta_i) = \delta_i [\dot{\ell}(y_i | \mathbf{x}_i, \boldsymbol{\theta}) - E^w \{\dot{\ell}(y_i | \mathbf{x}_i, \boldsymbol{\theta}) | \mathbf{x}_i, y_i, \delta_i = 1\}], \quad (9)$$

where E^w is the expectation under the assumed working model. Since IPW and MSCLE converges to different parameter estimations, we assume that IPW and MSCLE converges to $\boldsymbol{\theta}_l$ and $\boldsymbol{\theta}_{ls}$ respectively, which means $\boldsymbol{\theta}_{ls}$ is the solution of the following estimation equation (Wang & Kim 2022)

$$E^t[\psi_{SM}(\boldsymbol{\theta}; \mathbf{x}, y, \delta)] = 0, \quad (10)$$

OR

$$\mathbf{E}^t \left\{ \mathbf{E}^t \left[\dot{\ell}(\boldsymbol{\theta}_{ls}; \mathbf{x}, y) \pi(\mathbf{x}, y; \boldsymbol{\theta}) \right] - \frac{\mathbf{E}^t [\pi(\mathbf{x}, y; \boldsymbol{\theta}) | \mathbf{x}]}{\mathbf{E}^w [\pi(\mathbf{x}, y; \boldsymbol{\theta}) | \mathbf{x}]} \mathbf{E}^w \left[\dot{\ell}(\boldsymbol{\theta}_{ls}; \mathbf{x}, y) \pi(\mathbf{x}, y; \boldsymbol{\theta}) \right] \right\} = 0, \quad (11)$$

where we use $\mathbf{E}^t$ to emphasize that the expectation is taken with respect to the true data distribution. Under the model misspecification, the variance- correlation matrixs of MSCLE should be different.

We expand the Taylor expression of the score function of MSCLE, resulting in that

$$0 = \frac{1}{\sqrt{N}} \sum_{i=1}^N \psi_{SM}(\boldsymbol{\theta}_{ls}; \mathbf{x}_i, y_i, \delta_i) + \left[\frac{1}{N} \sum_{i=1}^N \nabla_{\boldsymbol{\theta}} \psi_{SM}(\boldsymbol{\theta}_{ls}; \mathbf{x}_i, y_i, \delta_i) \right] \sqrt{N}(\hat{\boldsymbol{\theta}}_S - \boldsymbol{\theta}_{ls}) + R_N,$$

where

$$R_N = \frac{1}{2} \sqrt{N}(\hat{\boldsymbol{\theta}}_S - \boldsymbol{\theta}_{ls}) \left[\frac{1}{N} \sum_{i=1}^N \nabla_{\boldsymbol{\theta}}^2 \psi_{SM}(\boldsymbol{\theta}_{ls}; \mathbf{x}_i, y_i, \delta_i) \right] (\hat{\boldsymbol{\theta}}_S - \boldsymbol{\theta}_{ls})^\top + o_P(1).$$

With the Assumption 1(2), we can prove that

$$R_N \leq \frac{1}{N} d \|\hat{\boldsymbol{\theta}}_S - \boldsymbol{\theta}_{ls}\| = o_P(1).$$

Also we have

$$B_{SM} = \mathbb{V}(\psi_{SM}(\boldsymbol{\theta}; \mathbf{x}, y, \delta)^{\otimes 2}) = \mathbf{E}^t[\delta(\dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}) - E^w\{\dot{\ell}(y | \mathbf{x}, \boldsymbol{\theta}) | \mathbf{x}, y, \delta\})^{\otimes 2}], \quad (12)$$

where $A^{\otimes 2} = AA^\top$. Then with the Assumption 2(1), let $A_{SM}^{-1} B_{SM} A_{SM}^{-1} = \Sigma_{SM}^{-1}$

$$\sqrt{N}(\hat{\boldsymbol{\theta}}_S - \boldsymbol{\theta}_{ls}) \xrightarrow{d} N(0, \Sigma_{SM}^{-1}),$$

while for IPW, we assume that the the asymptotic property is that

$$\sqrt{N}(\hat{\boldsymbol{\theta}}_W - \boldsymbol{\theta}_l) \xrightarrow{d} N(0, \Sigma_W^{-1}),$$

where $\Sigma_W^{-1} = A_W^{-1} B_W A_W^{-1} = \mathbf{E}^t[\dot{\ell}^{\otimes 2}(\boldsymbol{\theta}; \cdot, \mathbf{x}, y)]^{-1} \mathbf{E} \left\{ \frac{\dot{\ell}^{\otimes 2}(\boldsymbol{\theta}; \cdot, \mathbf{x}, y)}{\pi(\mathbf{x}, y)} \right\} \mathbf{E}^t[\dot{\ell}^{\otimes 2}(\boldsymbol{\theta}; \cdot, \mathbf{x}, y)]^{-1}$ and $B_W = \mathbf{E} \left[\frac{\dot{\ell}^{\otimes 2}(\boldsymbol{\theta}; \mathbf{x}, y)}{\pi(\mathbf{x}, y)} \right]$.

Otherwise, if the model is misspecified, then as $n \rightarrow \infty$, $P(H_N \geq c_\alpha) \rightarrow 1$, and $\hat{\boldsymbol{\theta}}_N$ converges asymptotically to the IPW estimator. We prove the following lemma.

Lemma 2. *When the model is misspecified, and the difference between IPW and MSCLE is $O_P(1/\sqrt{N})$, then we have $\hat{\boldsymbol{\theta}}_N \xrightarrow{p} \hat{\boldsymbol{\theta}}_W$.*

Proof. We first assume that under the model misspecification, the IPW and MSCLE converge to different parameter value,

$$\begin{aligned} \sqrt{N}(\hat{\boldsymbol{\theta}}_W - \boldsymbol{\theta}_l) &\xrightarrow{d} Z_W \sim N(0, \Sigma_W^{-1}), \\ \sqrt{N}(\hat{\boldsymbol{\theta}}_S - \boldsymbol{\theta}_{ls}) &\xrightarrow{d} Z_S \sim N(0, \Sigma_{SM}^{-1}). \end{aligned} \quad (13)$$

We notice that when the difference $\hat{\boldsymbol{\theta}}_W - \hat{\boldsymbol{\theta}}_S$ has a order larger than $\frac{1}{\sqrt{N}}$, and the divergence of the Hausman-type statistic and the consistency of the pre-test estimator remain valid provided that the covariance estimator $\mathbb{COV}(\hat{\boldsymbol{\theta}}_W, \hat{\boldsymbol{\theta}}_S)$ is consistent and positive definite. Under these conditions, $\lim_{N \rightarrow \infty} H_N = N(\hat{\boldsymbol{\theta}}_S - \hat{\boldsymbol{\theta}}_W)' \hat{V}_H^{-1}(\hat{\boldsymbol{\theta}}_S - \hat{\boldsymbol{\theta}}_W) \xrightarrow{p} \infty$, which easily derives that $\mathbf{I}\{H_N \geq c_\alpha\} \xrightarrow{p} 1$. Thus we have $\hat{\boldsymbol{\theta}}_S \mathbf{I}\{H_N < c_\alpha\} \xrightarrow{p} 0$, and we have $\hat{\boldsymbol{\theta}}_N \xrightarrow{p} \hat{\boldsymbol{\theta}}_W$. $\square$

Leeb & Pötscher (2005) and Leeb & Pötscher (2008) presented a conclusion that it is difficult or nearly impossible to obtain the uniform asymptotic normality of $\hat{\boldsymbol{\theta}}_N$. No estimator can consistently estimate the unconditional distribution of the model-selected estimator, not even locally. One reason is that the asymptotic variances of $\hat{\boldsymbol{\theta}}_S$ and $\hat{\boldsymbol{\theta}}_W$ differ, and their centers slightly shift under local misspecification, the distribution of $\hat{\boldsymbol{\theta}}_N$ often becomes multimodal. When the test statistic H_N fluctuates around the critical value c_α , the realized sample is intermittently assigned to the “narrow peak” (associated with $\hat{\boldsymbol{\theta}}_S$) or the “broad peak” (associated with $\hat{\boldsymbol{\theta}}_W$). Besides, the selection process is intrinsically correlated with the estimators themselves, as they are derived from the same subsampled data. This dependency induces significant skewness and heavy-tailedness in the finite-sample distribution. Specifically, the tail probability of $\hat{\boldsymbol{\theta}}_N$ can be substantially larger than that of a standard normal distribution.

1.1 A model averaging estimator

The biggest problem of handling $\hat{\boldsymbol{\theta}}_N$ is that it is a uncontinues estimator at the point of threshold $H_N = c_\alpha$. Thus, we use the model averaging method instead, by adding a smooth weight into the estimator. The new estimator

$$\hat{\boldsymbol{\theta}}_{NA} = w(H_N)\hat{\boldsymbol{\theta}}_W + (1 - w(H_N))\hat{\boldsymbol{\theta}}_S,$$

where $w(H_N)$ is a weight function related with H_N . We have three kinds options to choose for $w(\cdot)$.

- AIC criteria: $w(H_N) = \mathbf{I}\{H_N \geq c_\alpha\}$, which brings $\hat{\boldsymbol{\theta}}_{NA} = \hat{\boldsymbol{\theta}}_N$;
- Sigmoid Smoothing: $w(H_N) = \frac{1}{1 + \exp\{-(H_N - c_\alpha)/s\}}$, where s is the smoothing parameter.
- Adaptive weighting: $w(H_N) = \begin{cases} 0, & \text{if } H_N \leq c_\alpha, \\ 1 - \frac{c_\alpha}{H_N}, & \text{if } H_N > c_\alpha. \end{cases}$

Assumption 2. (1) $w(\cdot)$ is a measurable function with the value region of $[0, 1]$. $w(\cdot)$ should be almost continuous function so that when $D_N \xrightarrow{d} D$, we have $w(D_N) \xrightarrow{d} w(D)$.

For the assumption (1), since we commonly consider $w(\cdot)$ as sigmoid or huber function, the condition is satisfied naturely.

As we discussed, IPW and MSCLE may converges to different value $\boldsymbol{\theta}_l$ and $\boldsymbol{\theta}_{ls}$. We first consider the difference between $\boldsymbol{\theta}_l$ and $\boldsymbol{\theta}_{ls}$. When $\boldsymbol{\theta}_l - \boldsymbol{\theta}_{ls} = h$ where h is a constant, then we have $\lim_{N \rightarrow \infty} \hat{\boldsymbol{\theta}}_W - \hat{\boldsymbol{\theta}}_S = h$. As a result, $\lim_{N \rightarrow \infty} H_N \rightarrow \infty$. Let $\Delta = \hat{\boldsymbol{\theta}}_{NA} - \boldsymbol{\theta}_l$ and

$$Z_W = \hat{\boldsymbol{\theta}}_W - \boldsymbol{\theta}_l, \quad Z_S = \hat{\boldsymbol{\theta}}_S - \boldsymbol{\theta}_l,$$

as the Therome 4.1 in (Hjort & Claeskens 2003) claimed, we calculate the MSE $R(h)$ to evaluate the variance of the estimator $\hat{\boldsymbol{\theta}}_{NA}$ and the MSE is defined as

$$\begin{aligned} R(h) &= \text{tr}(\mathbf{E}(\Delta\Delta^\top)) = \text{tr}(\mathbf{E}[(w(H_N)Z_W + (1 - w(H_N))Z_S)^\top (w(H_N)Z_W + (1 - w(H_N))Z_S)]) \\ &= (1 - w(H_N))^2 \|h\|^2 + \frac{1}{N} \text{tr}(w(H_N)^2 \Sigma_W^{-1} + w(H_N)(1 - w(H_N))\Sigma_{SM}^{-1} + \text{tr}(C_{WS})) \\ &\xrightarrow{N \rightarrow \infty} (1 - w(H_N))^2 \|h\|^2, \end{aligned} \tag{14} \quad \{\text{eq:Rh}\}$$

where C_{WS} can be considered as $\text{COV}(Z_W, Z_S) = \text{COV}(\hat{\boldsymbol{\theta}}_W, \hat{\boldsymbol{\theta}}_S) = A_W^{-1} B_{WS} A_S^{-1}$. As long as $\lim_{N \rightarrow \infty} w(H_N) = 1$, then we can garuantee that $\lim_{N \rightarrow \infty} R(h) = 0$, while most of the $w(H_N)$ cases follows this requirement. Next we prove that $\boldsymbol{\theta}_{NA}$ is an unbiased estimator through

$$\lim_{N \rightarrow \infty} \mathbf{E}(\hat{\boldsymbol{\theta}}_{NA} - \boldsymbol{\theta}_l) = \lim_{N \rightarrow \infty} \mathbf{E}[(w(H_N)Z_W + (1 - w(H_N))Z_S)] \rightarrow \lim_{N \rightarrow \infty} \mathbf{E}[w(H_N)Z_W] = 0. \tag{15}$$

We next consider that the different between $\boldsymbol{\theta}_l$ and $\boldsymbol{\theta}_{ls}$ decreases as N increases. We assume that $\|\boldsymbol{\theta}_l - \boldsymbol{\theta}_{ls}\| \leq \frac{M}{\sqrt{N}}$, which is bounded by a value with order of $N^{-1/2}$ and $\delta_N = \sqrt{N}(\boldsymbol{\theta}_l - \boldsymbol{\theta}_{ls})$ which is a bounded sequence. As a result, through Bolzano-Weierstrass Theorem, there exists a subsequence $\{N_k\}$ such that when $k \rightarrow \infty$, δ_{N_k} converges to a constant vector δ_s and $\|\delta_s\| \leq M$. Thus let $D_N = \sqrt{N}(\hat{\boldsymbol{\theta}}_S - \hat{\boldsymbol{\theta}}_W)$. As a result,

$$\sqrt{N_k}(\hat{\boldsymbol{\theta}}_{W, N_k} - \hat{\boldsymbol{\theta}}_{S, N_k}) \xrightarrow{d} D \sim \mathbf{N}(\delta_s, V_H), \tag{16}$$

where $V_H = \Sigma_W^{-1} + \Sigma_{SM}^{-1} - C_{WS} - C_{WS}^\top$ is the limitation correlation matrix of $\hat{\boldsymbol{\theta}}_S - \hat{\boldsymbol{\theta}}_W$. The Hausman statistic can be formulated as

$$H_N = D_N \hat{V}_H^{-1} D_N^\top \xrightarrow{p} D V_H^{-1} D^\top \xrightarrow{d} \chi_\lambda^2,$$

while the center value $\lambda = \delta_s^\top V_H^{-1} \delta_s$ that can be bounded that

$$\delta_s^\top V_H^{-1} \delta_s \leq \lambda_{\max}(V_H^{-1}) M^2,$$

where $\lambda_{\max}(V_H^{-1})$ is the maximum eigenvalue of V_H^{-1} .

On the other hand, the difference between estimator $\hat{\boldsymbol{\theta}}_{NA}$ and $\boldsymbol{\theta}_l$ can be formulated as

$$\begin{aligned}\Delta &= \sqrt{N}(\hat{\boldsymbol{\theta}}_{NA} - \boldsymbol{\theta}_l) \\ &= \sqrt{N}[(w(H_N)(\hat{\boldsymbol{\theta}}_W - \boldsymbol{\theta}_l) + (1 - w(H_N))(\hat{\boldsymbol{\theta}}_S - \boldsymbol{\theta}_l))] = Z_W - (1 - w(H_N))(Z_W - Z_S + \delta_N),\end{aligned}\tag{17}$$

where

$$Z_W \sim \mathbf{N}(0, \Sigma_W^{-1}), \quad Z_W - Z_S \sim \mathbf{N}(\delta_s, V_H).$$

As long as $w(H_N)$ is bounded, based on the consistency of IPW and MSCLE, $\hat{\boldsymbol{\theta}}_{NA}$ also converges to $\boldsymbol{\theta}_0$. Similar with the conclusion of Lemma 2, we just have to study the property when the difference order is $O_P(1/\sqrt{N})$. We involve the concept of frequentist model average (FMA), which established a unified framework to address the problems of model selection and model averaging.

To evaluate the efficiency of $R(h)$, we next want to prove that $R(h)$ is bounded with respect to h , so that we can illustrate that the mixture estimator is superior than MSCLE. Then we have the following lemma:

Lemma 3.

$$\begin{aligned}\lim_{N \rightarrow \infty} R &= \text{tr}(\Sigma_W^{-1}) + \|\delta_s\|^2 \text{tr}(\mathbf{E}[(1 - w(\chi_{p+2}^2(\lambda)))^2] V_H) + \|\delta_s\|^2 \mathbf{E}[(1 - w(\chi_{p+4}^2(\lambda)))^2] \\ &\quad + \|\delta_s\|^2 \mathbf{E}[(1 - w(\chi_{p+2}^2(\lambda)))] \text{tr}((C_{WD} + C_{WD}^\top)) \\ &\quad + \|\delta_s\|^2 \{\mathbf{E}[(1 - w(\chi_{p+4}^2(\lambda)))^2] - \mathbf{E}[(1 - w(\chi_{p+2}^2(\lambda)))]\} \text{tr}((C_{WD} + C_{WD}^\top) V_H^{-1}).\end{aligned}\tag{18}$$

Should I prove that $R \leq \text{tr}(\Sigma_W^{-1})$ for all N

Proof. We first calculate the limit of $w(\chi_{p+2}^2(\lambda))$. We already know that $\frac{\chi^2(\lambda)}{\lambda} \xrightarrow{p} 1$ as $\lambda \rightarrow \infty$ and when $\delta_s \rightarrow \infty$, we have $\lim_{\delta_s \rightarrow \infty} \lambda = \lim_{\delta_s \rightarrow \infty} \delta_s V_H^{-1} \delta_s^\top \xrightarrow{p} \infty$. Plus, since $w(\lambda) \in [0, 1]$, thus we have $\lim_{\lambda \rightarrow \infty} w(\chi^2(\lambda)) \rightarrow 1$. Based on the dominated convergence theorem, we have $\lim_{h \rightarrow \infty} \mathbf{E}[w(\chi_{p+2}^2(\lambda))] \xrightarrow{p} 1$.

Assume that $D_h = V_H^{-1/2}(D - h)$, then $H_N = \|D_h\|^2$ and $Q = V_H^{1/2} \mathbf{E}(w(\|D_h\|^2) \|D_h\|^2) V_H^{-1/2}$, where D_h follows a normal distribution with mean of $V_H^{-1/2} h$ and variance of $\mathbf{I}_p$. Based on the lemma of non-center chi-square distribution, we have

$$\mathbf{E}(w(\|D_h\|^2) D_h) = V_H^{-1/2} h \mathbf{E}(w(\chi_{p+2}^2(\lambda))),$$

and

$$\mathbf{E}(w(\|D_h\|^2) \|D_h\|^2) = \mathbf{E}(w(\chi_{p+2}^2(\lambda))) \mathbf{I}_p + \mathbf{E}(w(\chi_{p+4}^2(\lambda))) V_H^{-1/2} h h^\top V_H^{-1/2},$$

where λ is non-center parameter and $\lambda = h^\top V_H^{-1} h$ in our concept.

We now divide $R(\delta_s)$ into four items. The first item is

$$\lim_{N_k \rightarrow \infty} \mathbf{E}(Z_{W,N_k}^2) = \Sigma_{W,N_k}^{-1}, \quad (19)$$

which can be proved directly through the conclusion of IPW.

The second item is

$$\begin{aligned} & \lim_{N_k \rightarrow \infty} \mathbf{E}((1 - w(H_{N_k}))^2 (Z_{S,N_k} - Z_{W,N_k} - \delta_s)(Z_{S,N_k} - Z_{W,N_k} - \delta_s)^\top) \\ &= V_H^{1/2} (\mathbf{E}[(1 - w(\chi_{p+2}^2(\lambda)))^2] \mathbf{I}_p + \mathbf{E}[(1 - w(\chi_{p+4}^2(\lambda)))^2] V_H^{-1/2} \delta_s \delta_s^\top V_H^{-1/2}) V_H^{1/2} \\ &= \mathbf{E}[(1 - w(\chi_{p+2}^2(\lambda)))^2] V_H + \mathbf{E}[(1 - w(\chi_{p+4}^2(\lambda)))^2] \|\delta_s\|^2. \end{aligned}$$

For the third item, we have to prove that

$$\begin{aligned} & \lim_{N_k \rightarrow \infty} \mathbf{E}[(1 - w(H_{N_k})) Z_{W,N_k} (Z_{S,N_k} - Z_{W,N_k} - \delta_s)^\top] \\ &= \mathbf{E}[(1 - w(\|D_{\delta_s}\|^2)) C_{WD} V_H^{-1} (D_{\delta_s} + \delta_s) D_{\delta_s}^\top], \end{aligned} \quad (20)$$

We first consider the joint distribution of Z_W and D_h , we have

$$\begin{pmatrix} Z_W \\ D_h \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ -h \end{pmatrix}, \begin{pmatrix} \Sigma_W^{-1} & C_{WD} \\ C_{WD}^\top & V_H \end{pmatrix} \right), \quad (21)$$

where $C_{WD} = \Sigma_W^{-1} - C_{WS}$. Then based on the conditional property of multinormal distribution, we can turn Z_W to $Z_W = C_{WD} V_H^{-1} (D_h + h) + \epsilon$, where ϵ is a standard normal variable which is independence with D . As a result, the third item in Eq. ?? can be formulated as

$$\begin{aligned} & \mathbf{E}(w(H_N)(Z_W(Z_S - Z_W - h)^\top)) \\ &= C_{WD} V_H^{-1} \mathbf{E}[w(\|D_h\|^2)(D_h + h) D_h^\top] \\ &= C_{WD} V_H^{-1} \mathbf{E}[w(\|D_h\|^2)(\|D_h\|^2 + h D_h^\top)] \\ &= C_{WD} V_H^{-1} [V_H \mathbf{E}[(1 - w(\chi_{p+2}^2(\lambda)))^2] \mathbf{I}_p + \mathbf{E}[(1 - w(\chi_{p+4}^2(\lambda)))^2] h h^\top] \\ &\quad - C_{WD} V_H^{-1} [h h^\top \mathbf{E}[(1 - w(\chi_{p+2}^2(\lambda)))^2]] \\ &= C_{WD} \mathbf{E}[(1 - w(\chi_{p+2}^2(\lambda)))^2] + C_{WD} V_H^{-1} h h^\top [\mathbf{E}[(1 - w(\chi_{p+4}^2(\lambda)))^2] - E[(1 - w(\chi_{p+2}^2(\lambda)))^2]]. \end{aligned} \quad (22)$$

Thus, we complete the proof. $\square$

Besides, we can also proof that when $\|\delta_s\|$ goes to infinity, the MSE of $\hat{\theta}_{NA}$ still converges to the one of IPW, showing the robustness of such estimator. For the second item, since $1 - w(\chi_{p+2}^2(\lambda)) \xrightarrow{p} 0$, then $V_H^{1/2} (\mathbf{E}[(1 - w(\chi_{p+2}^2(\lambda)))^2] \mathbf{I}_p) \rightarrow 0$. We also have

$$V_H^{1/2} \mathbf{E}[(1 - w(\chi_{p+4}^2(\lambda)))^2] V_H^{-1/2} \|\delta_s\|^2 V_H^{-1/2} V_H^{1/2} = \mathbf{E}[(1 - w(\chi_{p+4}^2(\lambda)))^2] \|\delta_s\|^2. \quad (23)$$

198 To ensure that this item converges to 0 and recall that $\lambda = \delta_s^\top V_H^{-1} \delta_s^\top \sim O_P(\|M\|^2)$, we just
 199 need to make sure that $w(H_N) \sim O_P(H_N^{-1})$. Sigmoid smoothing and adaptive weighting
 200 functions all satisfies this requirement. For the third item, we have

$$\lim_{\|\delta_s\| \rightarrow \infty} C_{WD} \mathbf{E}[(1 - w(\chi_{p+2}^2(\lambda)))^2] \rightarrow 0,$$

201 and

$$hh^\top [\mathbf{E}[(1 - w(\chi_{p+4}^2(\lambda)))^2] - \mathbf{E}[(1 - w(\chi_{p+2}^2(\lambda)))^2]] \rightarrow 0$$

202 when $\lambda = \delta_s^\top V_H^{-1} \delta_s \sim O_P(\|\delta_s\|^2)$. Then we prove that the third item also converges to 0, so
 203 as to the fourth part.

204 We assume that $w^*(H_N) = \arg \min_w R$. Thus, we have

205 **Lemma 4.** *We can obtain the optimal function w^* through*

$$w^*(H_N) = 1 - \frac{f_{\chi_{p+2}^2(\lambda)}(H_N) \text{tr}(C_{WD} + C_{WD}^\top) (\text{tr}(V_H^{-1}) - 1)}{2 \left[f_{\chi_{p+2}^2(\lambda)}(H_N) \text{tr}(V_H) + f_{\chi_{p+4}^2(\lambda)}(H_N) \text{tr}\{(C_{WD} + C_{WD}^\top) V_H^{-1}\} \right]}. \quad (24)$$

206 *Proof.* To minimize R , we differentiate R with respect to $w(H_N)$, then we have the following
 207 equation.

$$\begin{aligned} & -2(1 - w(H_N)) [f_{\chi_{p+2}^2(\lambda)}(H_N) \text{tr}(V_H) + f_{\chi_{p+4}^2(\lambda)}(H_N) \text{tr}\{(C_{WD} + C_{WD}^\top) V_H^{-1}\}] \\ & = f_{\chi_{p+2}^2(\lambda)}(H_N) \text{tr}(C_{WD} + C_{WD}^\top) (\text{tr}(V_H^{-1}) - 1). \end{aligned} \quad (25) \quad \{\text{R_differ}\}$$

208 By solving Eq. 25, we can obtain the optimized $w^*(H_N)$. □

209 On the other hand, the MSE of MSCLE

$$R_S(h) = N \mathbf{E}[(\hat{\beta}_S - \theta_l)(\hat{\beta}_S - \theta_l)^\top]^\top \quad (26)$$

210 , while we have

$$\sqrt{N}(\hat{\beta}_S - \theta_l) = \sqrt{N}(\hat{\beta}_S - \theta_{ls}) + \sqrt{N}(\hat{\theta}_S - \theta_l).$$

211 Thus

$$R_S(h) = \mathbf{E}[(Z_s - h)(Z_s - h)^\top] = \mathbf{E}[Z_s Z_s^\top] + hh^\top = V_S + hh^\top.$$

212 Then as $\|h\| \rightarrow \infty$, $R_S(h) \rightarrow \infty$.

213 References

214 Hausman, J. A. (1978), ‘Specification tests in econometrics’, *Econometrica: Journal of the*
 215 *econometric society* pp. 1251–1271.

- 216 Hjort, N. L. & Claeskens, G. (2003), ‘Frequentist model average estimators’, *Journal of the*
217 *American Statistical Association* **98**(464), 879–899.
- 218 Leeb, H. & Pötscher, B. M. (2005), ‘Model selection and inference: Facts and fiction’, *Econo-*
219 *metric Theory* **21**(1), 21–59.
- 220 Leeb, H. & Pötscher, B. M. (2008), ‘Can one estimate the unconditional distribution of
221 post-model-selection estimators?’, *Econometric Theory* **24**(2), 338–376.
- 222 Liu, L., Hudgens, M. G. & Becker-Dreps, S. (2016), ‘On inverse probability-weighted estima-
223 tors in the presence of interference’, *Biometrika* **103**(4), 829–842.
- 224 Van der Vaart, A. W. (2000), *Asymptotic statistics*, Vol. 3, Cambridge university press.
- 225 Wang, H. & Kim, J. K. (2022), ‘Maximum sampled conditional likelihood for informative
226 subsampling’, *Journal of machine learning research* **23**(332), 1–50.
- 227 White, H. (1982), ‘Maximum likelihood estimation of misspecified models’, *Econometrica:*
228 *Journal of the econometric society* pp. 1–25.